

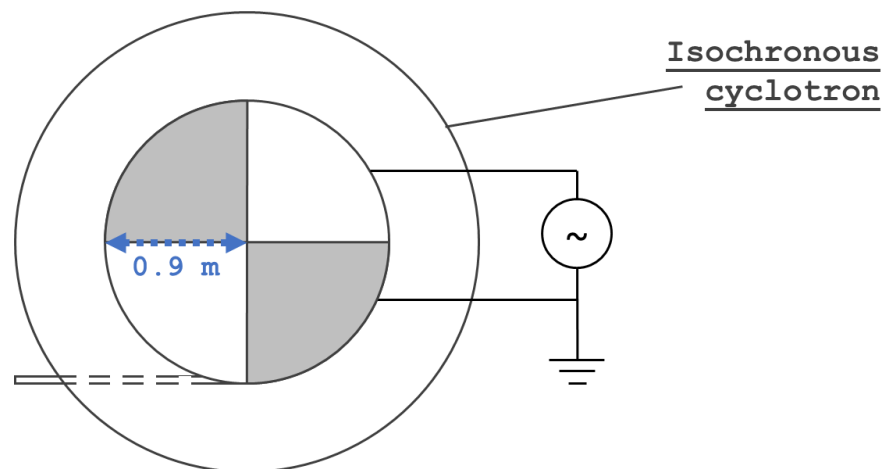
LGBIO2070 - Lab 2: Proton therapy system

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Context

The project aims to design a proton therapy system for lung cancer patients. A proton accelerator, beamline components, and other equipment are available for the project. Your task is to define the main parameters of the system (including beam production, beam transport, and beam scanning) and propose some risk mitigation.

1. Beam production



The proton accelerator is an isochronous cyclotron with 2 dees of 90° .

The acceleration cavity has a radius of 90 cm.

The proton source can provide up to 10 nA of beam current.

The objectives/constraints on the accelerator are that:

- The output proton energy is 200 MeV.
- The acceleration time is 0.9 ms.

The parameters you must determine to fulfill the objectives are:

- The magnetic field at injection (i.e. center) and extraction (i.e. maximum radius)
- The cyclotron frequency and the frequency of the high-voltage
- The high voltage amplitude

2. Beam transport

The distance between the output of the cyclotron and the isocenter where the patients will be treated is 7 m. A vacuum tube of 25 mm radius is installed between the two points. You have 4 quadrupoles to perform the beam optic along the line. The beam size (1 sigma) at the output of the cyclotron is 3 mm, and the divergence is 1 mm/m in both vertical and horizontal directions. The beam is diverging, and its emittance is equal to 0.

The objectives and constraints are:

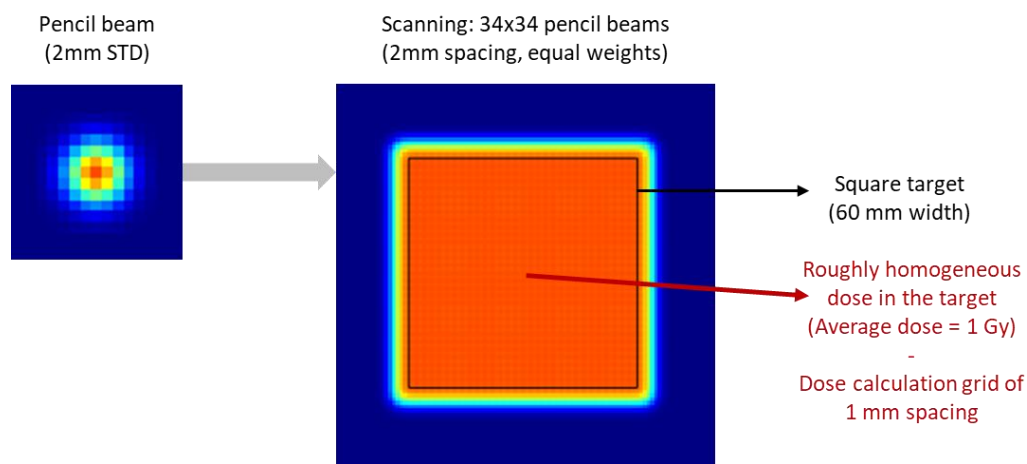
- The beam must be symmetrical at the isocenter (identical sizes in horizontal and vertical directions).
- The beam size (1 sigma) must be a maximum of 3 mm at the isocenter.
- The full beam size (3 sigma) must always stay smaller than the radius of the vacuum tube so that the protons do not interact with the walls of the tube anywhere along the beamline.

Propose a beamline configuration to fulfill the objectives. The results must include:

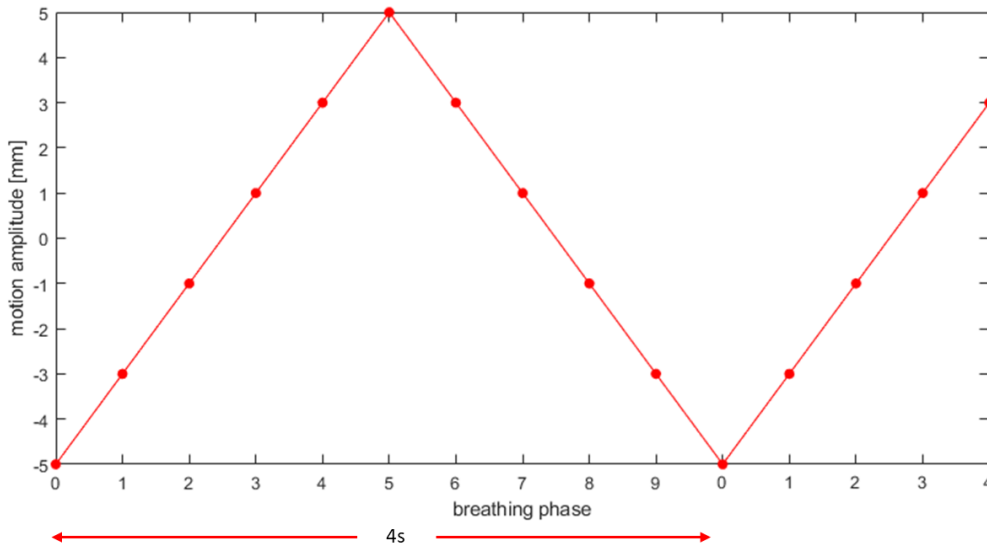
- The position and focusing direction of the quadrupoles along the beamline.
- The focal length of each quadrupole.
- A representation of the beam in phase-space at the different locations of interest along the beamline (for both horizontal and vertical directions).

3. Beam delivery

A 2D artificial test case is designed to validate treatment delivery at the isocenter. The target is a square of 60 mm width, and the treatment plan comprises 34x34 pencil beams. The pencil beam size (1 sigma) is 2 mm, and the spacing between pencil beams is 2 mm. All pencil beams have the same energy and deliver the same dose. The dose per pencil beam must be determined such that the average dose in the static target equals a prescribed dose of **1 Gy**. The scanning goes line after line, always in the same direction (i.e. no 'serpent-like' scanning). The scanning speed is such that the time between two consecutive pencil beams is **22 ms** (and the spot delivery is assumed to be instantaneous).



As the system is expected to treat mobile tumors, evaluating and mitigating interplay effects is required. The objective is that at least **95%** of the pixels in the target receive more than **90%** of the prescribed dose (i.e. 0.9 Gy). A horizontal periodic motion is introduced in this 2D artificial test case. The motion is a periodic triangle wave. Motion amplitude is **1 cm** and motion period is **4s**. The full motion period is divided into **10 phases** for interplay computation, as depicted in the following figure.



First, evaluate the impact of interplay on the dose delivered to the target when scanning lines are parallel to the motion and when scanning lines are orthogonal to the motion. In each case, if the objective is not reached, propose and demonstrate the efficiency of interplay reduction strategies to reach the objective. For every situation, display the resulting dose distribution.

Tips for Python implementation.

```
# Create the spot template
s_size = int(5 * spot_sigma + 1) # [pixel = mm]
spot = np.outer(np.exp(-0.5 * (np.arange(s_size) - (s_size - 1) / 2) ** 2 / spot_sigma ** 2),
                np.exp(-0.5 * (np.arange(s_size) - (s_size - 1) / 2) ** 2 / spot_sigma ** 2))

# Create a dose distribution from the list of xy spot positions
dose = np.zeros((100, 100))
for i in range(xy.shape[0]):
    dose[int(xy[i, 0] - (s_size - 1) / 2):int(xy[i, 0] + (s_size - 1) / 2 + 1),
        int(xy[i, 1] - (s_size - 1) / 2):int(xy[i, 1] + (s_size - 1) / 2 + 1)] += spot
```

4. Treatment safety

Finally, a technical file must demonstrate the general safety of the treatment delivery for moving targets. Identify the main risks associated with using your interplay reduction strategy to treat lung cancer patients (i.e. “what could possibly go wrong?”). Briefly describe a potential mitigation strategy for each identified risk, to ensure patient safety.